Geospatial proposal quality

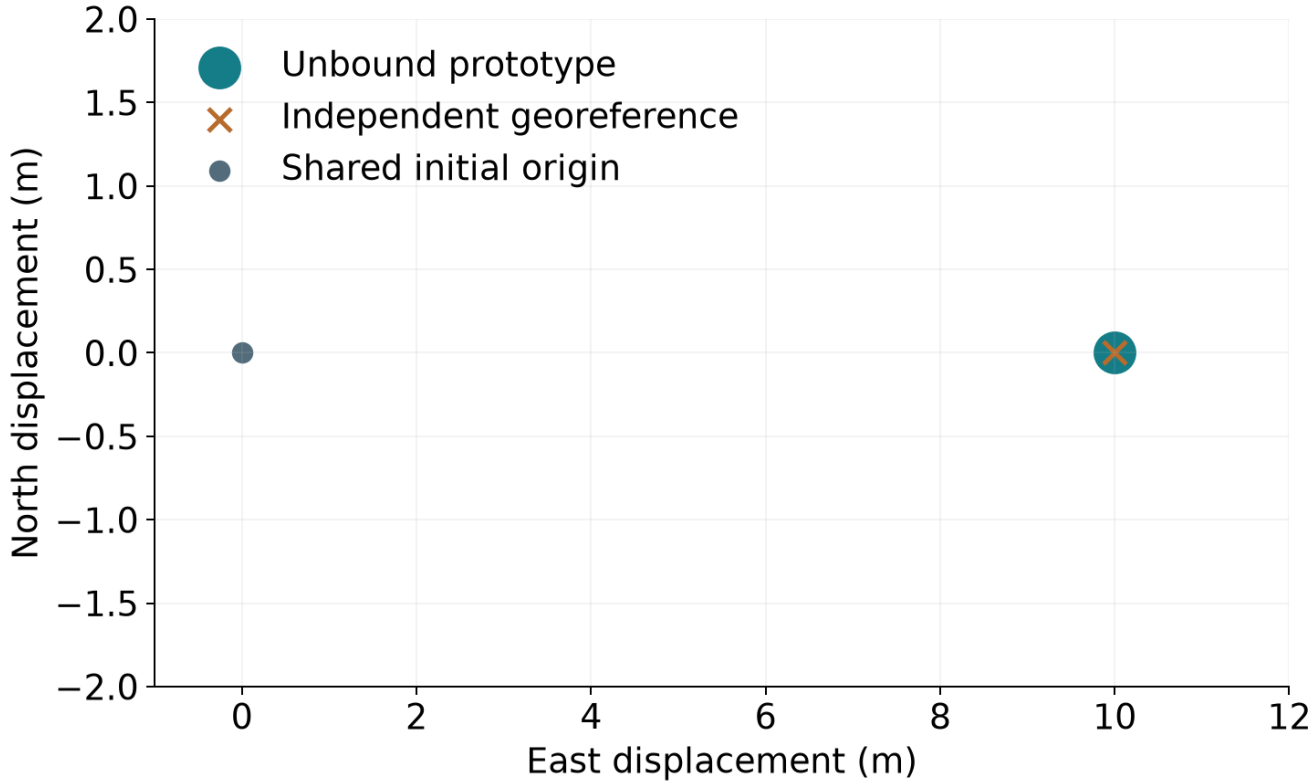
USD holds a model's shape. A coordinate reference system (CRS) explains a position on Earth.

The revised draft spells out instance movement, ordinary USD bake reuse and dependency coverage.

Fresh evidence supports the reviewed rules. Three prototype limits remain visible. These local leans still need group review.

Instance offsets retain one meaning

A local +X offset moves both coincident prototypes ten metres east, whether each has its own georeference or inherits one.



The draft specifies finite transport through the instancer frame. A required singular inverse fails.

Ordinary USD bake reassembly

Intent	Ordinary USD X (m)	Python / C++ X (m)
+100 m parent	6,378,237	6,378,237
Authored reset	6,378,137	6,378,137

A retained CRS identifies the Cartesian context. Ordinary USD determines transform inheritance and resets.

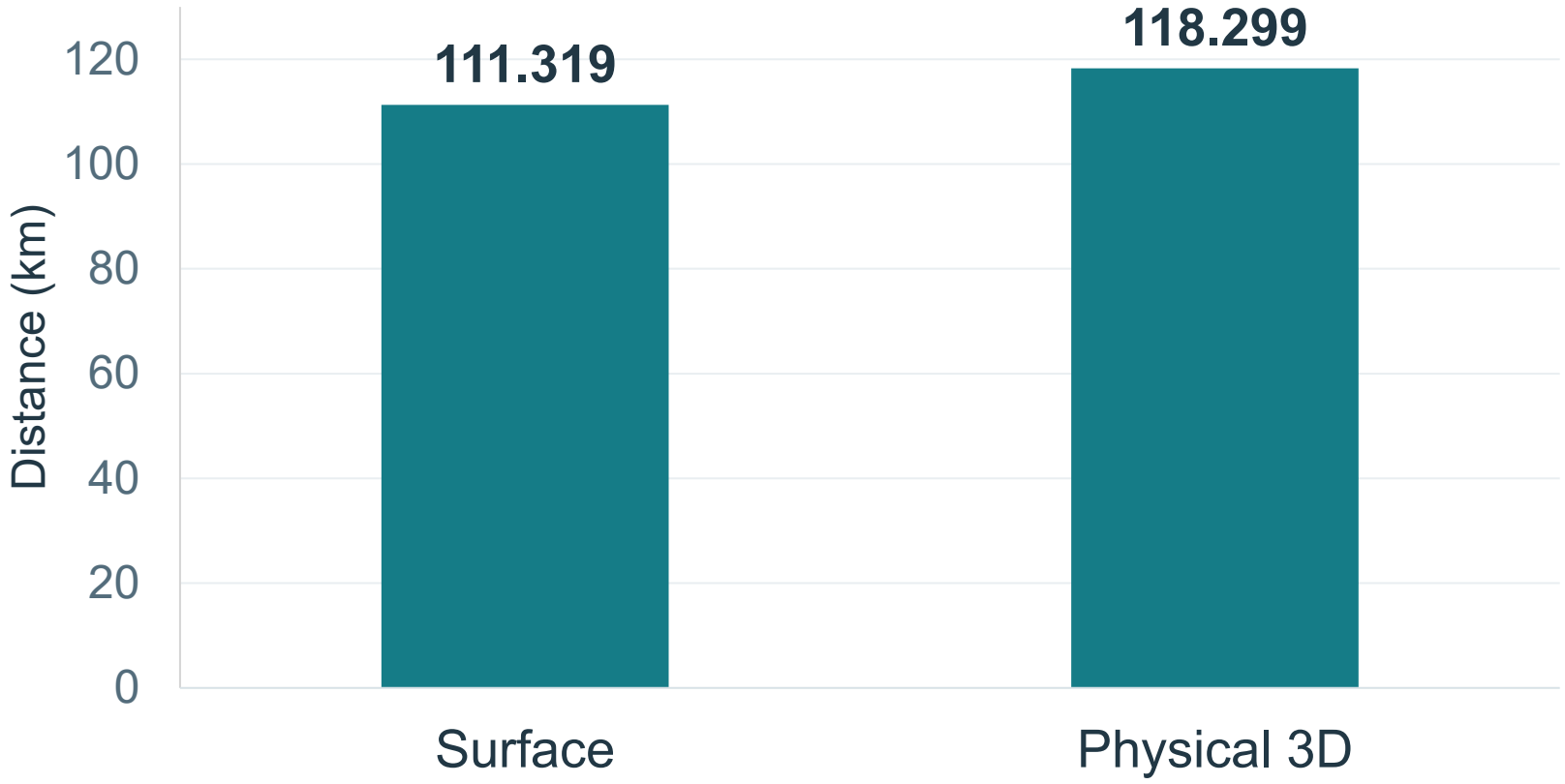
Geographic adjustment context

Request	Proposed outcome
Read geographic positions	Retain coordinates and native values
Adjust in an explicit Cartesian project frame	Apply the recorded ordinary adjustment
Adjust without a Cartesian project frame	Report unsupported adjustment

This limits one adjustment convention. Global data and analytical queries remain supported.

Physical 3D and surface distance

Two equatorial positions, one degree apart, both at 400 km ellipsoidal height.



The draft separates physical 3D separation from horizontal surface distance. Height remains explicit.

An authored rotation path

Authored poses	Resolved outcome
0 and +180 or -180 degrees	Ambiguous path fails
0, intermediate 90, then 180 degrees	Intermediate pose determines the path
Held reader or exact sample	Recorded pose remains defined

The source stores heading, pitch and roll. The reader computes a pose. The prototype uncertainty bound still needs evidence.

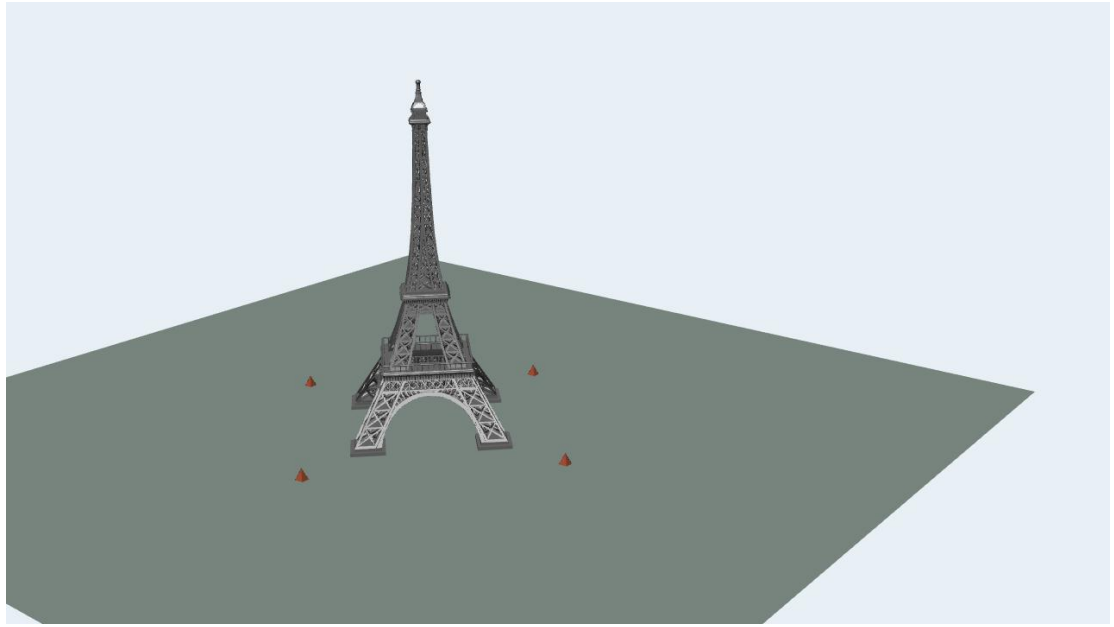
Dependency coverage with Profiles

Publisher declaration	Consumer interpretation
Complete summary, no retained use	Known none
Hard capability usage	Resolution dependency
Missing complete qualification	Unknown, conservatively hard

Existing Profiles metadata carries a proposed publishing profile. Publisher checks must cover retained variants and unloaded content. A metadata read alone cannot establish truth.

Working native visualization

Fresh native Hydra/Storm images of C++-resolved ordinary USD bakes.



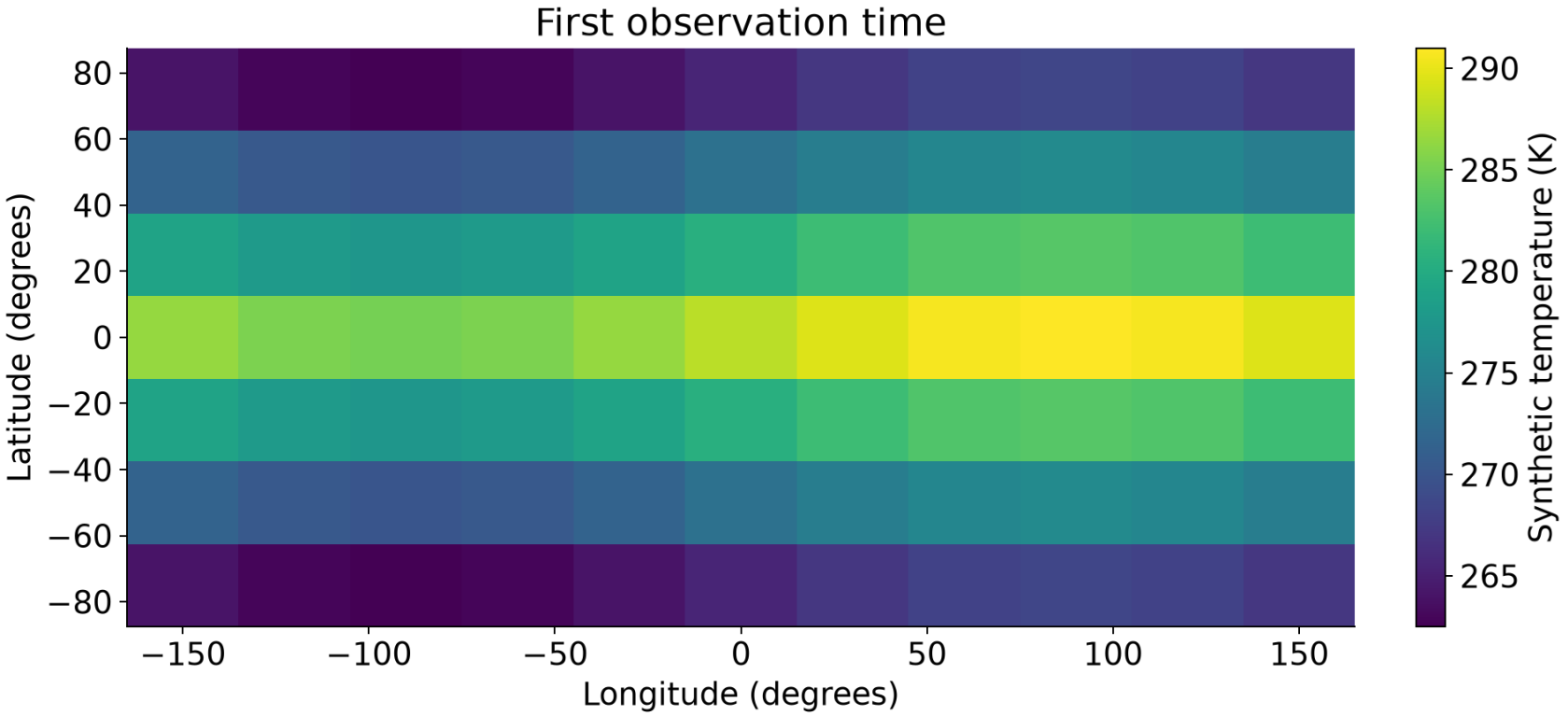
Eiffel model

Terrain

They demonstrate the render path. Surface-normal accuracy remains a separate failed diagnostic.

Measurements retain native meaning

The synthetic global field retains 154 observations over two times, with explicit height.



Geographic and Earth-centred results keep the same measurements. Named-dimension controls check the shared decoder.

Both readers resolve partner controls

France: CC49 to Lambert-93 (m)

Component	Python	C++
East	661,101.109417	661,101.109417
North	6,857,834.256536	6,857,834.256536
Height	37.298000	37.298000

Colorado: State Plane to site grid (US survey ft)

Component	Python	C++
East	3,108,871.450101	3,108,871.450101
North	1,206,317.469403	1,206,317.469403
Height	5,484.677949	5,484.677949

Recorded components in the declared output units. Both placement readers share PROJ and external-data decoding.

Remaining prototype limits

Finding	Assessment
Surface lighting directions	1.24 degrees across one large projected face. Current sampling lacks an error bound.
Near-half-turn numerical guard	Correct failure, incomplete rotational uncertainty evidence
WKT quoted text	Both readers mistake a quoted name for a deferred epoch wrapper
Publication and consumer coverage	Selected controls pass. Broader composition and continuous coverage remains unverified

The proposal supplies rules for these outcomes. Passing prototype tests cannot establish exhaustive proposal quality.

Review choices and fresh evidence

Evidence	Result
Contract tests	113 passed
Coordinates compared per reader	213,243
Prior cases	No unexpected numerical regression
Omniverse writeback	177,872 vertices and 48 edit checks

Review instance transport, Cartesian adjustment context and the publishing-profile contract. HPR, the half-turn policy and comparison conventions remain proposed.